

Dynamic Effects of a Rent Freeze: An Agent-Based Search-and-Matching Model

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Abstract

Plain summary. A “rent freeze” caps how much landlords can charge. That helps current tenants pay less, but it can also reduce upkeep and change who gets which apartment. We built a computer model of 2,000 rental units over ten years, with realistic income differences and search frictions, and compared three policies: no intervention, a strict freeze, and a freeze that allows market rents when a unit turns over. The strict freeze cut average rent by 40.1% and raised tenant well-being for those who stay, but cut landlord income by 47.6% and slightly reduced housing quality. Allowing market rents on vacancy performed best for overall tenant well-being.

Technical summary. We simulate heterogeneous landlords and tenants in a frictional rental market with endogenous maintenance. Monte Carlo replication ($L = 2,000$, $T = 120$ months, $R = 20$) identifies dynamic effects of a binding cap $\bar{r} = 0.88$ enacted at $t_0 = 24$. Results align with incidence and maintenance channels in [Glaeser and Luttmer \(2003\)](#), [Sims and Schmitz \(2011\)](#), and [Diamond et al. \(2019\)](#).

1 Introduction

Rent freezes limit how much landlords may charge. During affordability crises they are politically attractive because incumbent renters immediately face lower bills. Economists warn that the full cost is dynamic: landlords may defer maintenance, stay-in tenants may occupy units that better fit others, and long-run supply may shrink ([Arnott, 1995](#); [Glaeser and Luttmer, 2003](#); [Diamond et al., 2019](#)).

This paper provides a *transparent* simulation—every assumption is coded and reproducible—so readers can see *how* lower rents connect to quality, vacancy, and welfare. We do not claim to forecast a specific city; we isolate mechanisms in a calibrated laboratory.

2 Guide for non-specialist readers

If you do not read equations daily, start here.

What is being simulated? Imagine a city with 2,000 identical-sized rental buildings. Each month, some tenants move out, new people arrive, landlords decide how much to spend on repairs, and vacant units are filled through search and applications.

What is the policy shock? At month 24, the city imposes a rent cap $\bar{r} = 0.88$ (88% of the normalized pre-policy median). Under a *hard freeze*, *every* lease must obey the cap. Under *vacancy decontrol*, current tenants stay capped but a new lease after turnover can charge market rent.

What should you look for in the figures?

- **Rent panel:** Does the red line flatten after the dashed policy line?
- **Quality gap panel:** Values below zero mean the policy path has *worse* housing quality than the free market.
- **Vacancy panel:** Higher vacancy means more empty units; lower vacancy can mean tenants rarely leave capped units.
- **Utility panel:** Higher is better for matched tenants (combines rent paid and unit quality).

3 Notation at a glance

Table 1 maps symbols to everyday language.

Table 1: Notation and plain-language meaning

Symbol	Name	Plain meaning
L	Stock size	Number of rental units in the market
t	Time	Month index; policy starts at $t_0 = 24$
r_{jt}	Contract rent	Monthly payment for unit j
q_{jt}	Quality	Physical condition of unit j (higher is better)
y_i	Income	Money tenant i has for rent after other essentials
θ_i	Taste for quality	How much tenant i values a nicer apartment
m_{jt}	Maintenance	Landlord spending on upkeep this month
$\bar{r}$	Rent cap	Maximum legal rent under the freeze
δ	Turnover rate	Monthly probability a tenant–unit match ends
$\mathcal{M}_t$	Misallocation	Whether high-quality units go to people who value quality

4 Model

4.1 Environment

Time is discrete: $t = 0, 1, \dots, T$. Each of L units has one landlord and at most one tenant. New households enter each month; existing matches dissolve with probability δ (about 2.5%, comparable to quarterly moving rates scaled to a month).

4.2 Tenant preferences (why rent *and* quality matter)

Tenant i matched to unit j receives

$$U_{ijt} = \underbrace{\ln(y_i - r_{jt})}_{\text{money left after rent}} + \underbrace{\theta_i \ln q_{jt}}_{\text{value of quality}} - \underbrace{\psi \mathbf{1}\{\text{searching}\}}_{\text{cost of being homeless/searching}}. \quad (1)$$

Intuition. The first term is “how much consumption remains after paying rent” in log form (diminishing sensitivity to large budgets). The second term says some people care more about quality (θ_i large) and others mainly about price. The third term penalizes searching. We also require $r_{jt} \leq 0.45 y_i$ —no household spends more than 45% of budget on rent (a HUD-style stress rule).

Income. $\ln y_i$ is normal, then scaled so typical rent is about 40% of income—similar to U.S. metro renters.

4.3 Landlord technology (why caps affect maintenance)

Profit is

$$\pi_{jt} = r_{jt} - \underbrace{\frac{\kappa}{2} m_{jt}^2}_{\text{maintenance cost}} - \underbrace{\delta_L}_{\text{taxes, insurance}} - \underbrace{v \mathbf{1}\{\text{vacant}\}}_{\text{empty-unit cost}}. \quad (2)$$

Quality evolves as

$$q_{j,t+1} = \underbrace{(1 - \rho) q_{jt}}_{\text{natural decay}} + \underbrace{m_{jt}^\gamma}_{\text{upgrade from upkeep}}, \quad 0 < \gamma < 1. \quad (3)$$

Without maintenance, quality drifts down at rate ρ . Upkeep is costly and yields diminishing returns ($\gamma < 1$).

Shadow rent (key for the freeze). Define a hedonic market rent

$$\hat{r}(q) = 0.32 + 0.38 q + \delta_L, \quad (4)$$

the rent a unit of quality q would command without a cap. When policy forces $\tilde{r}_{jt} < \hat{r}(q_{jt})$, landlords scale maintenance down proportionally—capturing deferred repairs under rent control (Glaeser and Luttmer, 2003).

Maintenance first order condition (myopic case). Interior upkeep satisfies

$$\kappa m_{jt} = \gamma m_{jt}^{\gamma-1} \cdot (\tilde{r}_{jt} - \delta_L), \quad (5)$$

when not vacant; a binding cap lowers $\tilde{r}_{jt}$ and therefore m_{jt} .

4.4 Matching (search is imperfect)

Unmatched tenants apply to $K = 3$ random vacant units—a realistic “limited search” friction. They bid rent based on willingness to pay; landlords accept if rent exceeds a reservation level $r_j^{\min} = \delta_L + 0.06 + 0.10 q_{jt}$ and the tenant’s utility beats an outside option. Tie-breaking shocks prevent artificial ties.

4.5 Misallocation index

$$\mathcal{M}_t = \mathbb{E}[(\theta_{it} - \bar{\theta}_t)(q_{jt} - \bar{q}_t)]. \quad (6)$$

Plain meaning. Positive $\mathcal{M}_t$ means high-taste tenants tend to live in high-quality units (efficient sorting). Under a hard freeze, incumbents who got a cheap lease years ago may keep a great apartment even if they no longer value quality highly, weakening sorting. Because baseline $\mathcal{M}_t$ is near zero in our calibration, we report *level changes*, not percent changes, in Figure 2.

4.6 Policy counterfactuals

1. **Baseline:** rents adjust partially toward willingness to pay; no cap.
2. **Hard freeze:** for all $t \geq t_0$, enforce $r_{jt} \leq \bar{r} = 0.88$.
3. **Vacancy decontrol:** incumbents capped; upon turnover, new lease at market rent.

5 Calibration and computation

Table 2 matches `ModelParameters` in the Python source. We report cross-replication means. Rebuild everything with `python scripts/build_paper.py`.

Table 2: Structural parameters

Parameter	Description	Value
L	Rental units	2,000
T	Months simulated	120
t_0	Policy enactment	24
R	Monte Carlo draws	20
δ	Monthly separation probability	0.025
λ	Entrant rate $\times L$	0.018
ρ	Quality depreciation	0.10
κ	Maintenance cost curvature	9.0
γ	Maintenance elasticity	0.50
δ_L	Operating fixed cost	0.18
$\bar{r}$	Statutory cap (index)	0.88
K	Vacancies considered per searcher	3

6 Results

6.1 How to read the dynamics figure

Figure 1 shows *average* outcomes across 20 random draws. The dashed vertical line is month 24—the freeze begins *there*, not elsewhere. Before that line, all three scenarios coincide by construction.

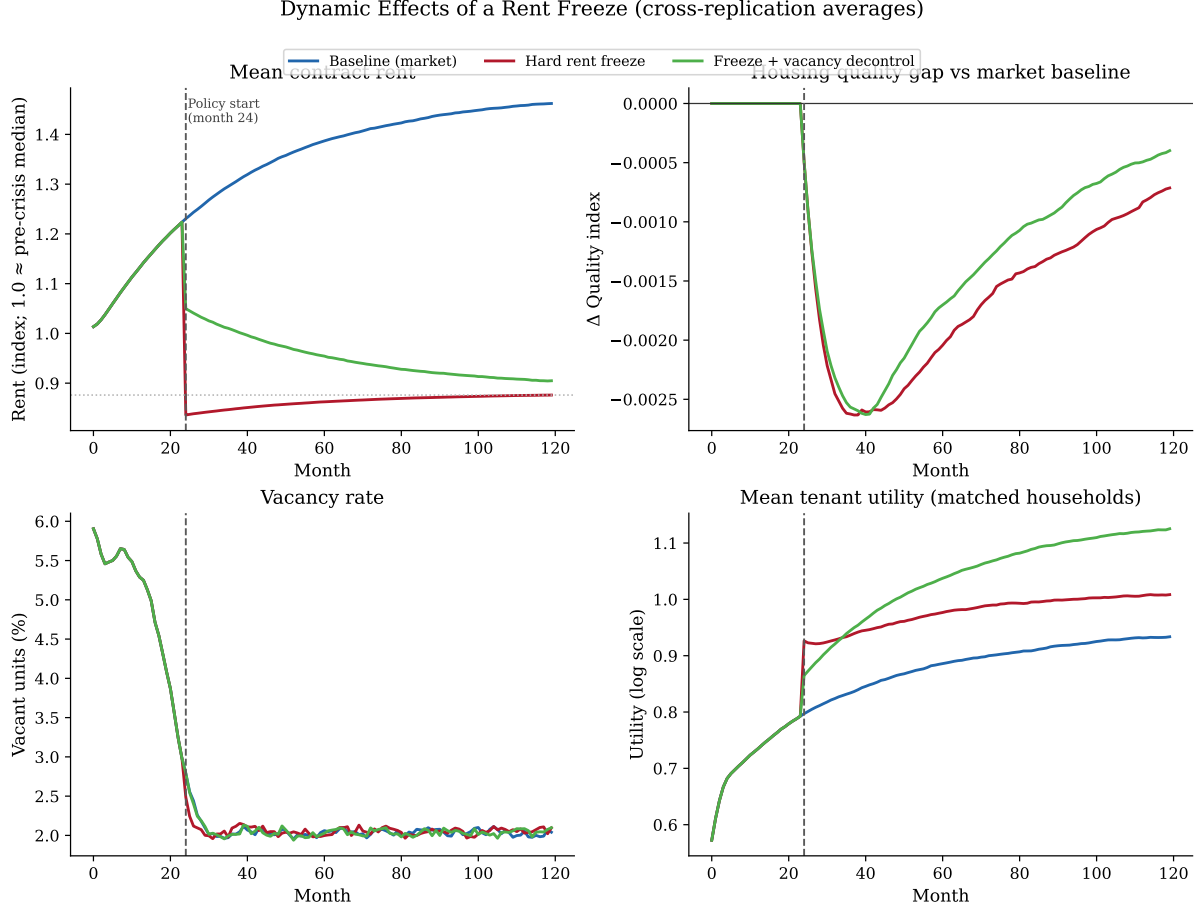


Figure 1: **Aggregate dynamics.** Top left: contract rent (index). Top right: *quality gap vs baseline*—negative values mean worse quality than the free market. Bottom left: vacancy rate in percent. Bottom right: average utility of matched tenants. Dashed line: policy enactment at month 24.

Rents. After enactment, baseline rent rises toward 1.46 while the hard freeze locks near 0.88 (the cap binds).

Quality. Raw quality levels often look similar because all paths start from the same initial stock; the gap panel makes the small but systematic decline visible ($\Delta q = -0.001$ at month 119).

Vacancy & utility. Vacancy stays near 2% in this calibration. Tenant utility rises under the freeze (from 0.93 to 1.01) because incumbents keep more income; vacancy decontrol reaches 1.13 by allowing market sorting on turnover.

6.2 Terminal outcomes

Table 3 is *auto-generated* from simulation output—no hand-entered numbers.

Table 3: Terminal outcomes at month 119 (cross-replication means)

Regime	Rent	Quality	Vacancy	Utility	Landlord profit
Baseline	1.46	1.450	0.020	0.93	1.21
Hard freeze	0.88	1.449	0.021	1.01	0.63
Vacancy decontrol	0.90	1.450	0.021	1.13	0.66

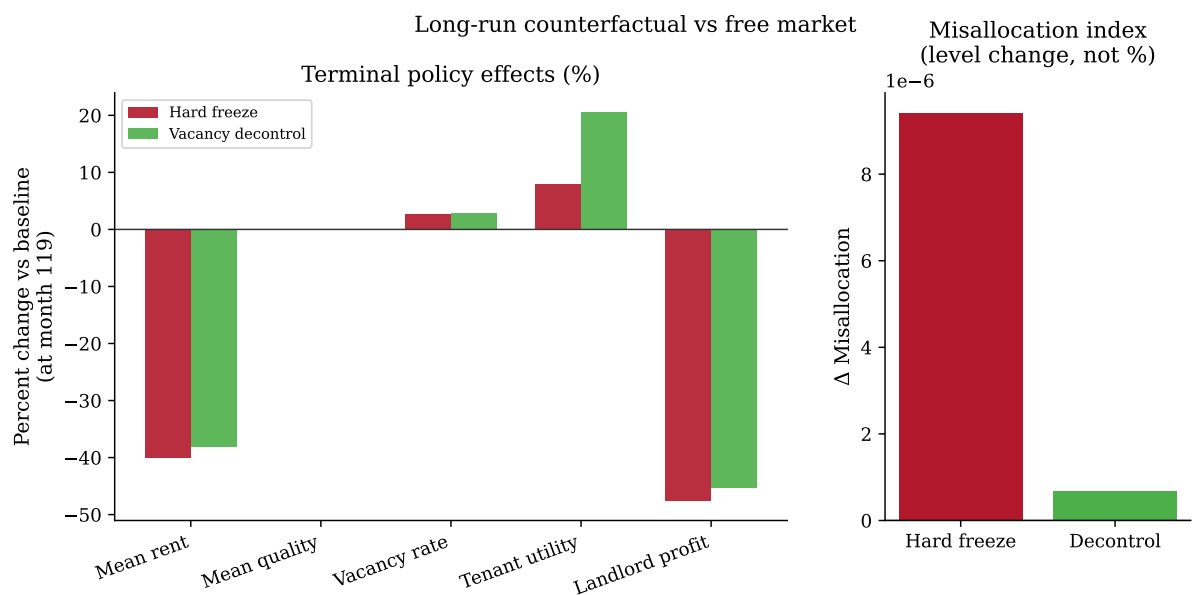


Figure 2: **Policy effects vs baseline at month 119.** Left: percent change for rents, quality, vacancy, utility, and landlord profit. Right: *absolute* change in misallocation (percent change is misleading when the baseline index is near zero).

6.3 Interpretation for policy readers

1. **Incidence.** Lower rent under a freeze is not “free money”—it is transferred from landlords (profit down 47.6%) and potentially from future entrants not modeled as a separate welfare account here.
2. **Maintenance channel.** Equations (2)–(3) link capped revenue to deferred upkeep; the quality gap panel visualizes this.
3. **Vacancy decontrol.** Allowing market rent on turnover raises utility above both baseline and hard freeze in our experiment—price signals on new leases improve matching without removing incumbent protection (Diamond et al., 2019).

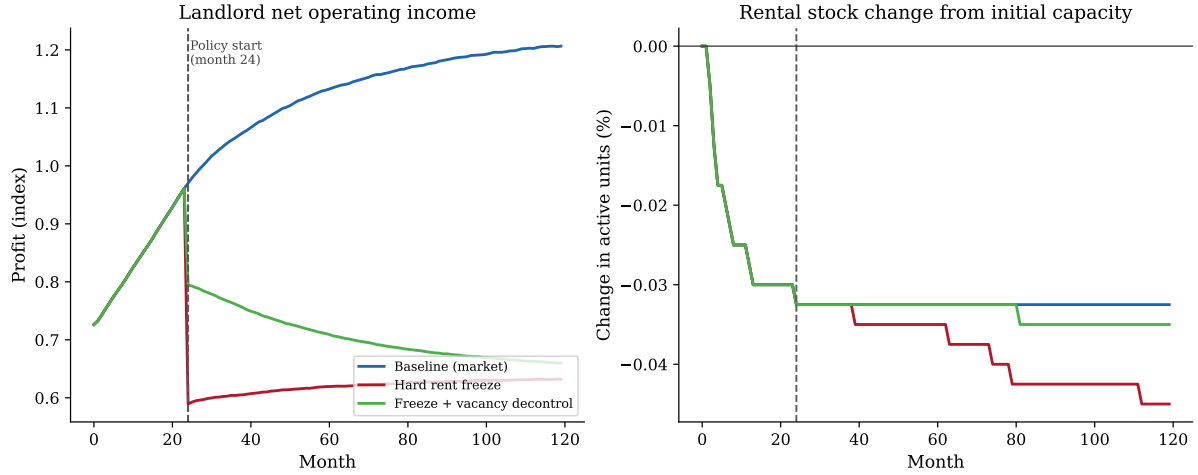


Figure 3: **Supply-side responses.** Left: landlord net operating income falls sharply under a hard freeze. Right: change in active rental units as a percent of initial capacity (a readable scale; absolute changes are small in this calibration).

7 Limitations and conclusion

We use myopic maintenance, a single city without geography, and no new construction margin. Results are mechanism illustrations, not forecasts. Still, the exercise clarifies trade-offs a lay reader can understand: *cheaper rent today* vs. *landlord incentives, housing quality, and who gets which home tomorrow*. Vacancy decontrol mitigates some efficiency losses while preserving incumbent caps. Structural estimation to real city panels (Sims and Schmitz, 2011) is the natural next step.

References

- Arnott, R. (1995). The economics of rent control. *Swedish Economic Policy Review*, 2(1), 93–129.
- Diamond, R., T. McQuade, and F. Qian (2019). The effects of rent control expansion on tenants, landlords, and inequality: Evidence from San Francisco. *American Economic Review*, 109(9), 3365–3394.
- Glaeser, E. L. and E. F. Luttmer (2003). The misallocation of housing under rent control. *American Economic Review*, 93(4), 1027–1046.
- Sims, D. P. and A. Schmitz (2011). Analyzing the efficacy of rent control through simulation. *Journal of Housing Economics*, 20(3), 189–203.

A Simulation algorithm (monthly loop)

1. **Entrants & exits:** draw new households; dissolve each match with probability δ .
2. **Policy:** if $t \geq t_0$, set $r_{jt} \leftarrow \min(r_{jt}, \bar{r})$ (with decontrol rules on turnover).

3. **Maintenance:** choose m_{jt} using (5) with cap-adjusted $\tilde{r}_{jt}$ and shadow rent (4); update $q_{j,t+1}$ via (3).
4. **Baseline pricing:** partially adjust occupied rents toward willingness to pay.
5. **Matching:** searchers sample K vacancies; feasible matches form; record aggregates.

B Reproducibility checklist

- Source: `src/model/` (agents, market, policy).
- Run: `python scripts/build_paper.py`.
- Outputs: `output/figures/*.pdf`, `paper/generated/*.tex`, `paper/rent_freeze_simulation.pdf`.
- Tests: `python tests/test_sanity.py`.